

Equilibrium & fsolve

- Review so far
- fsolve
- Examples
- Full Method

~~HW3 due today 6pm~~~~• Q4 has issue w/upload.
I will try to fix it today.~~

HW4 due Friday @ 6pm

OH
1-3pm Upson 447

Review so farTopicsTechniques

EOM

- Particle Dynamics
- Zero-length spring
- Vector and Dyads
- Rotation Matrices
- Conservation Lin/Ang. Momentum.
- Energy & Work
- Equilibrium

• Animating $\left\{ \begin{array}{l} \text{drawnow} \\ \text{gif} \\ \text{videos} \end{array} \right.$

• ode45

• fsolve.

fsolve Solution to Non-Linear Equations
(need to DL Optimization Toolbox)

linear Eq^s

$$3a + 5b = 0$$

$$9a - 6b = 0$$

Eom $\dot{X} = f(X)$

$$\ddot{x} = -\frac{k(r-L_0)}{m} \frac{\vec{x}}{|\vec{r}|} - \frac{c v_r}{m} \frac{\vec{x}}{|\vec{r}|} = 0$$

Non-linear Eq^s

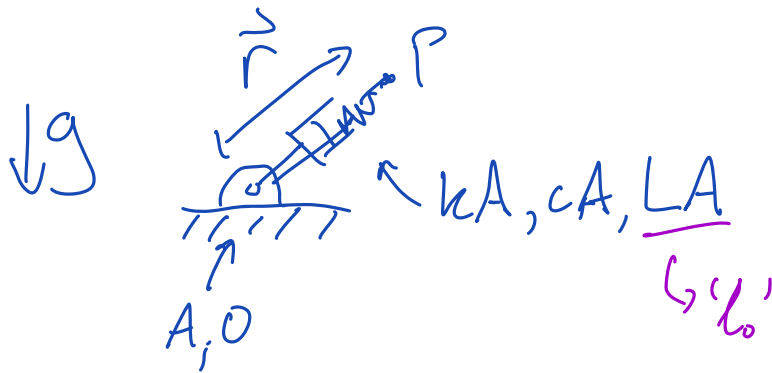
$$3e^a + 5ba = 0$$

$$6\frac{a}{b} - 9\cos(b) = 0$$

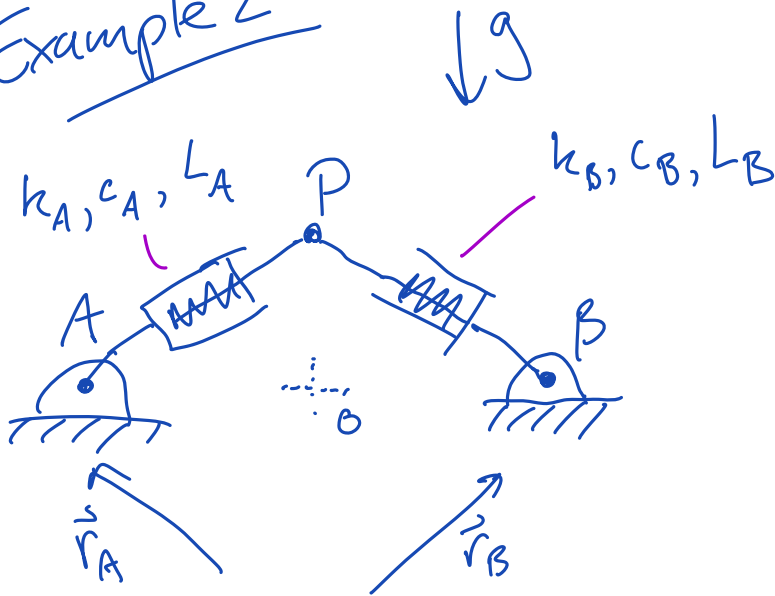
$$\ddot{y} = -\frac{k(r-L_0)}{m} \frac{\vec{y}}{|\vec{r}|} - \frac{c v_r}{m} \frac{\vec{y}}{|\vec{r}|} - g = 0$$



use fsolve
on $f(X)=0$

Example 1Matlab

- 1) Create $f(x)$
- 2) Search for roots of $f(x)=0$ using 'solve'
- 3) Evaluate Stability

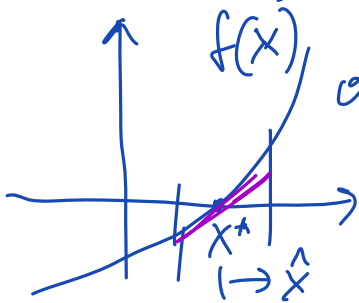
Example 2

- How many equilibrium pts?
- What is their stability?

Full Method

- 1) Create function $f(x)$ in Matlab
- 2) Look for equilibrium solutions such that $f(x) = 0$ using `fsolve`. We can call such a solution $X = X^*$.
- 3) Evaluate stability using one of two methods
 - a) Brute force: Try multiple ICs near to X^* and check the trend.

- b) Linearization: Derive the linear approximation of $f(x)$ in the vicinity of X^* .



$$\dot{\hat{X}} = A \hat{X}$$

Jacobian.

- A is a matrix of constants.
- $\hat{X} = X - X^*$

If any eigenvalue of A is positive, then the point is unstable.

fsolve for Periodic Solution

Normally we have EoMs, $\dot{X} = f(x)$

function $[Xdot] = rhs(t, X)$

to find $X = X(t)$

$$[t, X] = ode45(rhs, t, X0)$$

We want $X(t) = X(t + \Delta t)$, we need
function that outputs the final state, and
set

$$\underbrace{f(X(t)) - f(X(t + \Delta t))}_{fsolve} = 0$$

function

$$[Xend] = \text{final_state}(tend, X0) \quad \swarrow \text{ode45}$$

$$f_{periodic} = @X \quad rhs(t, X) - rhs(t, \text{final_state}(tend, X0))$$

$$fsolve(f_{periodic}, X0, options)$$

